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Manufacturing device of roof panel for vehicles

Abstract

A manufacturing apparatus of a roof panel includes a first mold, a second mold including first trimming units provided in a second lower die and a second upper die to trim an edge portion of a first processed plate processed by the first mold, second trimming units to trim a sunroof hole in the first processed plate, first restriking units to form first undercut faces on both edge portions of the sunroof hole, and first cam forming units to form a hinge mounting portion and a third mold including flange forming units provided in a third lower die and a third upper die to form a down flange on an edge portion of the sunroof hole of a second processed plate processed by the second mold, and second restriking units provided in the third lower die and the third upper die to form a water receiving portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) The present application claims priority to Korean Patent Application No. 10-2022-0090016 filed on Jul. 21, 2022, the entire contents of which is incorporated herein for all purposes by this reference.

BACKGROUND OF THE PRESENT DISCLOSURE

Field of the Present Disclosure

(2) The present disclosure relates to a manufacturing system of a vehicle body panel. More particularly, the present disclosure relates to a manufacturing device of a roof panel for a vehicle, which forms the roof panel to a predetermined shape in a sub-assembling line of a vehicle body.

Description of Related Art

(3) In general, it is necessary to go through several molding processes to produce one complete vehicle body panel.

(4) A vehicle body panel may be formed into a predetermined shape by drawing, trimming, flanging, restriking, and piercing processes of a panel material (e.g., blank).

(5) For example, a roof panel for a vehicle is manufactured through four molds (e.g., a first mold, a second mold, a third mold, and a fourth mold) in the related art.

(6) The first mold is used to perform the drawing process of forming the panel material into the predetermined shape. The second mold is used to perform the trimming process of cutting edges of the panel material. The third mold is used to perform a detailed trimming process and the restriking process of the panel material, and the fourth mold is used to perform the flanging process and the piercing process, etc., and finally mold the roof panel.

(7) However, when the number of molds increases to perform the above processes, not only the manufacturing cost of the mold increases, but also a factory site used for manufacturing the mold also increases.
(8) Furthermore, as the number of molds increases, a manufacturing period of the vehicle body panel also increases. Furthermore, when the number of molds increases, the time required for setting up the equipment in a mass production plate increases, which acts as an unfavorable factor for production management.
(9) Accordingly, there is a demand for research and development of a method capable of simplifying the number of molds and the number of processes required for manufacturing the body panel.
(10) The information disclosed in this Background of the present disclosure is only for enhancement of understanding of the general background of the present disclosure and may not be taken as an acknowledgement or any form of suggestion that this information forms the prior art already known to a person skilled in the art.
BRIEF SUMMARY
(11) Various aspects of the present disclosure are directed to providing a manufacturing apparatus of a roof panel for a vehicle capable of reducing the number of molds and the number of processes for forming the roof panel.
(12) A manufacturing apparatus of a roof panel for a vehicle according to various exemplary embodiments of the present disclosure may include i) a first mold in which a blank holder and a lower drawing steel are provided in a first lower die, and an upper drawing steel corresponding to the lower drawing steel is provided in a first upper die, to draw a blank in a predetermined shape, ii) a second mold including first trimming units provided in a second lower die and a second upper die to trim an edge portion of a first processed plate processed by the first mold, second trimming units provided on the second lower die and the second upper die to trim a sunroof hole in a front portion of the first processed plate, first restriking units provided on the second lower die and the second upper die to form first undercut faces on first and second edge portions of the sunroof hole in a left and right direction, and first cam forming units provided on the second lower die and the second upper die to form a hinge mounting portion on a rear portion of the first processed plate, and iii) a third mold including flange forming units provided in a third lower die and a third upper die to form a down flange on an edge portion of the sunroof hole of a second processed plate processed by the second mold, and second restriking units provided in the third lower die and the third upper die to form a water receiving portion connected to the hinge mounting portion on a rear portion of the second processed plate.
(13) A manufacturing apparatus of a roof panel for a vehicle according to various exemplary embodiments of the present disclosure may include i) a first mold in which a blank holder and a lower drawing steel are provided in a first lower die, and an upper drawing steel corresponding to the lower drawing steel is provided in a first upper die, to draw a blank in a predetermined shape, ii) a second mold including first trimming units provided in a second lower die and a second upper die to trim an edge portion of a first processed plate processed by the first mold, second trimming units provided on the second lower die and the second upper die to trim a sunroof hole in a front portion of the first processed plate, and first restriking units provided on the second lower die and the second upper die to form first undercut faces on first and second edge portions of the sunroof hole in a left and right direction, and iii) a third mold including flange forming units provided in a third lower die and a third upper die to form a down flange on an edge portion of the sunroof hole of a second processed plate processed by the second mold.
(14) A manufacturing apparatus of a roof panel for a vehicle according to various exemplary embodiments of the present disclosure may include i) a first mold in which a blank holder and a lower drawing steel are provided in a first lower die, and an upper drawing steel corresponding to the lower drawing steel is provided in a first upper die, to draw a blank in a predetermined shape,

ii) a second mold including first trimming units provided in a second lower die and a second upper die to trim an edge portion of a first processed plate processed by the first mold, and first cam forming units provided on the second lower die and the second upper die to form a hinge mounting portion on a rear portion of the first processed plate, and iii) a third mold including second restriking units provided in the third lower die and the third upper die to form a water receiving portion connected to the hinge mounting portion on a rear portion of a second processed plate processed by the second mold.

(15) Furthermore, in the manufacturing apparatus of the roof panel for the vehicle according to various exemplary embodiments of the present disclosure, the first trimming units may form at least one molding mounting portion on each of first and second edge portions of the first processed plate in the left and right direction thereof.

(16) Furthermore, in the manufacturing apparatus of the roof panel for the vehicle according to various exemplary embodiments of the present disclosure, the third mold may further include third trimming units provided on the third lower die and the third upper die to trim a rear end portion of the second processed plate; and a pair of cam trimming units provided on the third lower die and the third upper die to trim first and second sides of the rear end portion of the second processed plate, respectively.

(17) Furthermore, in the manufacturing apparatus of the roof panel for the vehicle according to various exemplary embodiments of the present disclosure, the third mold may trim the rear end portion of the second processed plate through the third trimming units before forming the water receiving portion through the second restriking units.

(18) Furthermore, in the manufacturing apparatus of the roof panel for the vehicle according to various exemplary embodiments of the present disclosure, the third mold may trim first and second sides of the rear end portion of the second processed plate, respectively, through the pair of cam trimming units after forming the water receiving portion through the second restriking units.

(19) Furthermore, in the manufacturing apparatus of the roof panel for the vehicle according to various exemplary embodiments of the present disclosure, the third mold may further include second cam forming units provided on the third lower die and the third upper die to form a second undercut face on at least one corner of the down flange.

(20) Furthermore, in the manufacturing apparatus of the roof panel for the vehicle according to various exemplary embodiments of the present disclosure, the third mold may further include third cam forming units provided on the third lower die and the third upper die, to form an up flange on the at least one molding mounting portion formed on the second processed plate.

(21) Furthermore, in the manufacturing apparatus of the roof panel for the vehicle according to various exemplary embodiments of the present disclosure, the first trimming units may include a first lower trimming steel provided on the second lower die, and a first upper trimming steel provided on the second upper die in correspondence to the first lower trimming steel.

(22) Furthermore, in the manufacturing apparatus of the roof panel for the vehicle according to various exemplary embodiments of the present disclosure, the second trimming units may include a second lower trimming steel connected to an internal edge portion of the first lower trimming steel, and a second upper trimming steel connected to an internal edge portion of the first upper trimming steel in correspondence to the second lower trimming steel, and including a pressing protrusion extending in a downward direction to support the first processed plate.

(23) Furthermore, in the manufacturing apparatus of the roof panel for the vehicle according to various exemplary embodiments of the present disclosure, the first restriking units may include first lower restriking steels connected to the first lower trimming steel and the second lower trimming steel and including first undercut forming grooves formed therein; and first upper restriking steels connected to the first upper trimming steel and the second upper trimming steel and including first undercut forming protrusions corresponding to the first undercut forming grooves formed thereon.

(24) Furthermore, in the manufacturing apparatus of the roof panel for the vehicle according to various exemplary embodiments of the present disclosure, the first cam forming units may include a swing cam rotatably provided on a die base mounted on the second lower die through a rotation center shaft in a front and rear direction and including a first lower cam forming steel mounted thereon, and a first cam slide slidably provided on the second upper die to be in cam contact with the swing cam, and including an upper cam forming steel corresponding to the first lower cam forming steel mounted thereon.
(25) Furthermore, in the manufacturing apparatus of the roof panel for the vehicle according to various exemplary embodiments of the present disclosure, the rotation center shaft may be combined to an eccentric point biased to a rear side from the center portion of the swing cam.
(26) Furthermore, in the manufacturing apparatus of the roof panel for the vehicle according to various exemplary embodiments of the present disclosure, the first cam forming units may further include a driving cylinder provided on the die base to be operable forwards and backwards in the front and rear direction, and a support block combined to an operation rod of the driving cylinder to support a front side of the swing cam.
(27) Furthermore, in the manufacturing apparatus of the roof panel for the vehicle according to various exemplary embodiments of the present disclosure, the flange forming units may include a lower flange forming steel including separating blocks respectively disposed on first and second sides thereof in the left and right directions, and provided on the third lower die, and an upper flange forming steel provided in the third upper die in correspondence to the lower flange forming steel, and including through holes formed in first and second corners of the rear thereof.
(28) Furthermore, in the manufacturing apparatus of the roof panel for the vehicle according to various exemplary embodiments of the present disclosure, each of the separating blocks may include a second undercut forming groove connected to each of the through holes.
(29) Furthermore, in the manufacturing apparatus of the roof panel for the vehicle according to various exemplary embodiments of the present disclosure, the second cam forming units may include a second cam slide including a second lower cam forming steel including second undercut forming protrusions combined to the second undercut forming grooves through the through holes, the second cam slide being slidably provided on the third lower die, and first cam drives provided on the third lower die and the third upper die to slide and move the second cam slide.
(30) Furthermore, in the manufacturing apparatus of the roof panel for the vehicle according to various exemplary embodiments of the present disclosure, the third cam forming units may include a third cam slide combined to each of the separating blocks and slidably provided on the third lower die, a plurality of seesaw cams fixed to first and second sides of the third lower die in the left and right direction, mounted on the third cam slide, and each including a third lower cam forming steel provided thereon, a second cam drive provided on the third upper die to be in cam contact with the third cam slide, and a plurality of third cam drives provided on the third upper die to be in cam contact with each of the seesaw cams.
(31) Furthermore, in the manufacturing apparatus of the roof panel for the vehicle according to various exemplary embodiments of the present disclosure, the third mold may further include an upper pad movably provided on the third upper die in a vertical direction through a plurality of gas springs.
(32) Furthermore, in the manufacturing apparatus of the roof panel for the vehicle according to various exemplary embodiments of the present disclosure, the third trimming units may include a third lower trimming steel provided on the third lower die, and a third upper trimming steel provided on the upper pad in correspondence to the third lower trimming steel.
(33) Furthermore, in the manufacturing apparatus of the roof panel for the vehicle according to various exemplary embodiments of the present disclosure, the second restriking units may include a second lower restriking steel connected to the third lower trimming steel, and a second upper restriking steel provided on the upper pad with a height difference from the third upper trimming

steel.

(34) Furthermore, in the manufacturing apparatus of the roof panel for the vehicle according to various exemplary embodiments of the present disclosure, each of the pair of cam trimming units may include a fourth cam slide slidably provided on the third upper die and including a cam trimming steel provided thereon, and a fourth cam drive provided on the third lower die to be in cam contact with the fourth cam slide.

(35) In various exemplary embodiments of the present disclosure, unlike the related art, because the roof panel may be manufactured through three molds, the number of molds and the number of processes may be reduced, and accordingly, the mold investment cost, the mold material cost, the processing cost, and the cost required for the production management, etc. may be reduced.

(36) Furthermore, the effects obtainable or predicted by various exemplary embodiments of the present disclosure are disclosed directly or implicitly in the detailed description of various exemplary embodiments of the present disclosure. That is, various effects predicted according to various exemplary embodiments of the present disclosure will be disclosed in the detailed description to be described below.

(37) The methods and apparatuses of the present disclosure have other features and advantages which will be apparent from or are set forth in more detail in the accompanying drawings, which are incorporated herein, and the following Detailed Description, which together serve to explain certain principles of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a block diagram schematically illustrating a manufacturing device of a roof panel for a vehicle according to various exemplary embodiments of the present disclosure.

(2) FIG. **2** and FIG. **3** are views exemplarily illustrating examples of a roof panel manufactured by a manufacturing device of a roof panel for a vehicle according to various exemplary embodiments of the present disclosure.

(3) FIG. **4** is a view exemplarily illustrating a first mold applied to a manufacturing device of a roof panel for a vehicle according to various exemplary embodiments of the present disclosure.

(4) FIG. **5** is a view exemplarily illustrating a first processed plate formed by a first mold applied to a manufacturing device of a roof panel for a vehicle according to various exemplary embodiments of the present disclosure.

(5) FIG. **6** is a view exemplarily illustrating a second mold applied to a manufacturing device of a roof panel for a vehicle according to various exemplary embodiments of the present disclosure.

(6) FIG. **7** is a view exemplarily illustrating a second processed plate formed by a second mold applied to a manufacturing device of a roof panel for a vehicle according to various exemplary embodiments of the present disclosure.

(7) FIG. **8** and FIG. **9** are views exemplarily illustrating a first trimming unit, a second trimming unit, and a first restriking unit of a second mold applied to a manufacturing device of a roof panel for a vehicle according to various exemplary embodiments of the present disclosure.

(8) FIG. **10**, FIG. **11**, and FIG. **12** are views exemplarily illustrating first cam forming units of a second mold applied to a manufacturing device of a roof panel for a vehicle according to various exemplary embodiments of the present disclosure.

(9) FIG. **13** is a view exemplarily illustrating a third mold applied to a manufacturing device of a roof panel for a vehicle according to various exemplary embodiments of the present disclosure.

(10) FIG. **14**, and FIG. **15** are views exemplarily illustrating a flange forming unit and a second cam forming unit of a third mold applied to a manufacturing device of a roof panel for a vehicle according to various exemplary embodiments of the present disclosure.

(11) FIG. **16** and FIG. **17** are views exemplarily illustrating a third cam forming unit of a third mold applied to a manufacturing device of a roof panel for a vehicle according to various exemplary embodiments of the present disclosure.
(12) FIG. **18**, FIG. **19** and FIG. **20** are views exemplarily illustrating a third forming unit, a second restriking unit, and a cam trimming unit of a third mold applied to a manufacturing device of a roof panel for a vehicle according to various exemplary embodiments of the present disclosure.
(13) It may be understood that the appended drawings are not necessarily to scale, presenting a somewhat simplified representation of various features illustrative of the basic principles of the present disclosure. The specific design features of the present disclosure as included herein, including, for example, specific dimensions, orientations, locations, and shapes will be determined in part by the particularly intended application and use environment.
(14) In the figures, reference numbers refer to a same or equivalent parts of the present disclosure throughout the several figures of the drawing.
DETAILED DESCRIPTION
(15) Reference will now be made in detail to various embodiments of the present disclosure(s), examples of which are illustrated in the accompanying drawings and described below. While the present disclosure(s) will be described in conjunction with exemplary embodiments of the present disclosure, it will be understood that the present description is not intended to limit the present disclosure(s) to those exemplary embodiments of the present disclosure. On the other hand, the present disclosure(s) is/are intended to cover not only the exemplary embodiments of the present disclosure, but also various alternatives, modifications, equivalents and other embodiments, which may be included within the spirit and scope of the present disclosure as defined by the appended claims.
(16) Exemplary embodiments of the present disclosure will be described more fully hereinafter with reference to the accompanying drawings, in which embodiments of the present disclosure are shown. However, the present disclosure may be embodied in several different forms, and is not limited to the exemplary embodiment described herein.
(17) The terms used herein are for describing specific embodiments, and are not intended to limit the present disclosure. As used herein, the singular forms are also intended to include the plural forms unless the context clearly dictates otherwise. As used herein, it should also be understood that the terms ‘comprise’ and/or ‘comprising’ indicate the presence of specified features, integers, steps, operations, elements and/or components, but do not exclude the presence or addition of one or more other features, integers, steps, etc. elements, operations, components, and/or groups thereof. As used herein, the term ‘combined’ indicates a physical relationship between two components in which the components are directly connected to each other or indirectly connected through one or more intervening components.
(18) Furthermore, as used herein, the term ‘and/or’ includes any one or all combinations of one or more of associated listed items, and ‘operably connected’ or similar terms mean that at least two members are directly or indirectly connected to each other to transmit power. However, two operatively connected members do not always rotate at the same speed and in the same direction.
(19) Furthermore, the terms “vehicle,” “of a vehicle,” “automobile,” or other similar terms used herein generally may include passenger automobiles including a passenger vehicle including a sports vehicle, a sports utility vehicle (SUV), a bus, a truck, and various commercial vehicles, and may include a hybrid vehicle, an electric vehicle, a hybrid electric vehicle, a hydrogen power vehicle, and other alternative fuel vehicles (e.g., fuels derived from resources other than petroleum).
(20) Hereinafter, embodiments of the present disclosure will be described in detail with reference to the accompanying drawings.
(21) FIG. **1** is a block diagram schematically illustrating a manufacturing device of a roof panel for a vehicle according to various exemplary embodiments of the present disclosure.

(22) Referring to FIG. **1**, a manufacturing device **100** of the roof panel for the vehicle according to various exemplary embodiments of the present disclosure may be applied to a press system that forms a vehicle body panel into a predetermined shape in a vehicle body sub-assembly line.
(23) The press system may include a drawing process, a trimming process, a flanging process, a restriking process, and a piercing process.
(24) In one example, the manufacturing device **100** of the roof panel for the vehicle according to various exemplary embodiments of the present disclosure may form a roof panel **70** to which a sunroof hole is applied into the predetermined shape through the above-described processes.
(25) In another example, the manufacturing device **100** of the roof panel for the vehicle according to various exemplary embodiments of the present disclosure may form the roof panel **70** to which a tailgate may be hinged into the predetermined shape through the above-described processes.
(26) Furthermore, the manufacturing device **100** of the roof panel for the vehicle according to various exemplary embodiments of the present disclosure may also form the roof panel **70** to which the sunroof hole and a hinge mounting structure of the tailgate are applied into the predetermined shape through the above-described processes.
(27) However, hereinafter, an example of forming the roof panel **70** to which the sunroof hole and the hinge mounting structure of the tailgate are applied into the predetermined shape will be described.
(28) In the present specification, a longitudinal direction of the roof panel **70** may be defined as a front and rear direction (e.g., a vehicle body longitudinal direction), and a width direction (e.g., a width direction of the vehicle) of the roof panel **70** may be defined as a left and right direction thereof.
(29) And, in the present specification, the ‘upper end portion’, ‘upper portion’, ‘upper end’ or ‘upper surface’ of the constituent element indicates the end portion, part, end, or surface of the constituent element at a relatively upper side in the drawing, and the ‘lower end portion’, ‘lower portion’, ‘lower end’ or ‘lower surface’ of the constituent element indicates the end portion, part, end, or surface of the constituent element at a relatively lower side in the drawing.
(30) Furthermore, in the present specification, an end (e.g., one side end or another side end, etc.) of the constituent element indicates an end point of the constituent element in any one direction, and an end portion (e.g., one side end portion or another one side end portion, etc.) of the constituent element indicates a certain part of the constituent element including an end point thereof.
(31) The manufacturing device **100** of the roof panel for the vehicle according to various exemplary embodiments of the present disclosure includes a structure configured for reducing the number of molds and the number of processes required for forming the roof panel **70**.
(32) To the present end, the manufacturing device **100** of the roof panel for the vehicle according to various exemplary embodiments of the present disclosure includes a first mold **110**, a second mold **210**, and a third mold **310**.
(33) In various exemplary embodiments of the present disclosure, the first mold **110** may form a blank **10** as a panel material into a predetermined shape. The second mold **210** may form a first processed plate **30** processed by the first mold **110** into a different shape, and the third mold **310** may form a second processed plate **50** processed by the second mold **210** into the final roof panel **70**.
(34) The configurations and operations of the first mold **110**, the second mold **210**, and the third mold **310** will be described in more detail below.
(35) Meanwhile, in one example, the roof panel **70** manufactured by the manufacturing device **100** of the roof panel for the vehicle according to various exemplary embodiments of the present disclosure includes a sunroof hole **71** formed in the front portion as shown in FIG. **2** and FIG. **3**.
(36) Furthermore, the roof panel **70** includes a down flange **73** extending in a downward direction from an edge portion of the sunroof hole **71** and at least one molding mounting portion **75**

respectively formed on both edge portions of the sunroof hole **71** in left and right direction thereof.
(37) Here, the sunroof hole **71** and the down flange **73** are configured to mount a sunroof glass. Furthermore, the at least one molding mounting portion **75** is configured to mount a roof side molding.
(38) Furthermore, the at least one molding mounting portion **75** includes an up flange **77**. Furthermore, the roof panel **70** includes a hinge mounting portion **79** formed on the rear portion and a water receiving portion **81** connected to the hinge mounting portion **79**.
(39) In the above, the up flange **77** is configured to mount a clip for fixing roof side molding to the at least one molding mounting portion **75**. The hinge mounting portion **79** is configured to mount a hinge of a tailgate, and the water receiving portion **81** is configured to induce water flowing along a glass surface of a sunroof glass to the side of the vehicle.
(40) Furthermore, the down flange **73** as described above includes first undercut faces **83** and second undercut faces **85** for fixing the sunroof glass.
(41) Here, the first undercut faces **83** may be formed in the down flange **73** at positions corresponding to both edge portions of the sunroof hole **71** in the front and rear direction thereof. Furthermore, the second undercut faces **85** may be formed on at least one corner of the down flange **73**. In one example, the second undercut faces **85** may be formed at the corners of both rear sides of the down flange **73**, respectively.
(42) Hereinafter, the configurations of the first mold **110**, the second mold **210**, and the third mold **310** as described above will be described in detail with reference to FIG. **1**, FIG. **2**, and FIG. **3** and the accompanying drawings.
(43) FIG. **4** is a view exemplarily illustrating a first mold applied to a manufacturing device of a roof panel for a vehicle according to various exemplary embodiments of the present disclosure.
(44) Referring to FIGS. **1** and **4**, in various exemplary embodiments of the present disclosure, the first mold **110** is configured to draw the blank **10** cut to a predetermined size into a predetermined shape. As shown in FIG. **5**, the first mold **110** may draw and form the blank **10** into a first processed plate **30** including a predetermined shape.
(45) The first mold **110** includes a first lower die **111**, a blank holder **113**, a lower drawing steel **115**, a first upper die **117**, and an upper drawing steel **119**.
(46) The first lower die **111** is fixed to the floor of a process workshop. The blank holder **113** is configured to support the edge portion of the blank **10** (hereinafter see FIG. **5**). The blank holder **113** is provided on the first lower die **111** to be movable in a vertical direction thereof. The blank holder **113** may be moved up and down through a plurality of cushion springs known to a person of an ordinary skill in the art. The lower drawing steel **115** (also referred to as 'punch' by a person of an ordinary skill in the art) is configured to form a lower surface of the first processed plate **30**. The lower drawing steel **115** is fixed to the upper portion of the first lower die **111**, and is disposed inside the edge portion of the blank holder **113**.
(47) The first upper die **117** is provided to be movable in the vertical direction in correspondence to the first lower die **111**. The first upper die **117** may reciprocate in the vertical direction by a hydraulic pressure cylinder device known to a person of an ordinary skill in the art. The upper drawing steel **119** is configured to form an upper surface of the first processed plate **30**. The upper drawing steel **119** is fixed to the lower portion of the first upper die **117** in correspondence to the lower drawing steel **115**.
(48) FIG. **6** is a view exemplarily illustrating a second mold applied to a manufacturing device of a roof panel for a vehicle according to various exemplary embodiments of the present disclosure.
(49) Referring to FIGS. **1** and **6**, in various exemplary embodiments of the present disclosure, the second mold **210** is configured to form the first processed plate **30** (hereinafter see FIG. **5**) processed by the first mold **110** into a predetermined shape.
(50) The second mold **210** may form the first processed plate **30** into a second processed plate **50** of the predetermined shape as shown in FIG. **7** by trimming, restriking, and cam forming the first

processed plate **30**.
(51) The second mold **210** includes a second lower die **211**, a second upper die **231**, a first trimming unit **221**, a second trimming unit **241**, first restriking units **261**, and first cam forming units **281**.
(52) The second lower die **211** is fixed to the floor of the process workshop. The second upper die **231** is provided to be movable in the vertical direction in correspondence to the second lower die **211**. The second upper die **231** may reciprocate in the vertical direction by a hydraulic pressure cylinder device known to a person of an ordinary skill in the art.
(53) In various exemplary embodiments of the present disclosure, the first trimming unit **221** is configured to trim the edge portion of the first processed plate **30** processed by the first mold **110**. The first trimming unit **221** may entirely trim the edge portion of the front portion of the first processed plate **30** and both edge portions thereof in the left and right direction thereof. Furthermore, the first trimming unit **221** may form at least one molding mounting portion **75** as shown in FIG. **7** at each of both edge portions of the first processed plate **30** in the left and right direction thereof. The first trimming unit **221** is provided on the second lower die **211** and the second upper die **231**.
(54) As shown in FIG. **6**, FIG. **7**, FIG. **8**, and FIG. **9**, the first trimming unit **221** includes a first lower trimming steel **223** and a first upper trimming steel **225**.
(55) As shown in FIG. **8**, the first lower trimming steel **223** is provided on an upper portion of the second lower die **211**. The first lower trimming steel **223** is configured to support the edge portion of the front portion of the first processed plate **30** and both edge portions in the left and right direction thereof.
(56) As shown in FIG. **9**, the first upper trimming steel **225** is provided on a lower portion of the second upper die **231** in correspondence to the first lower trimming steel **223**. The first upper trimming steel **225** is configured to cut the edge portion of the front portion of the first processed plate **30** placed on the first lower trimming steel **223** and both edge portions in the left and right direction thereof.
(57) Referring to FIG. **6**, FIG. **7**, FIG. **8**, and FIG. **9**, in various exemplary embodiments of the present disclosure, the second trimming unit **241** is configured to trim the sunroof hole **71** as shown in FIG. **7** in the front portion of the first processed plate **30**. In one example, the sunroof hole **71** may be a substantially quadrangle hole. The second trimming unit **241** is provided on the second lower die **211** and the second upper die **231**.
(58) The second trimming unit **241** includes a second lower trimming steel **243** and a second upper trimming steel **245**.
(59) As shown in FIG. **8**, the second lower trimming steel **243** is provided on an upper portion of the second lower die **211**. The second lower trimming steel **243** is configured to support the front portion of the first processed plate **30**. The second lower trimming steel **243** is connected to an internal edge portion of the first lower trimming steel **223**.
(60) As shown in FIG. **9**, the second upper trimming steel **245** is provided on a lower portion of the second upper die **231** in correspondence to the second lower trimming steel **243**. The second upper trimming steel **245** is configured to cut the front portion of the first processed plate **30** placed on the second lower trimming steel **243** in the shape of the sunroof hole **71**.
(61) The second upper trimming steel **245** is connected to an internal edge portion of the first upper trimming steel **225** in correspondence to the second lower trimming steel **243**. Furthermore, the second upper trimming steel **245** includes a pressing protrusion **247** extending in a downward direction thereof. The pressing protrusion **247** is configured to support and presses the first processed plate **30**.
(62) Referring to FIG. **6**, FIG. **7**, FIG. **8**, and FIG. **9**, in various exemplary embodiments of the present disclosure, the first restriking units **261** are configured to respectively form the first undercut faces **83** at both edge portions of the sunroof hole **71** in the left and right direction as

shown in FIG. **7**. The first restriking units **261** are respectively provided on the second lower die **211** and the second upper die **231**.
(63) The first restriking units **261** include first lower restriking steels **263** and first upper restriking steels **265**.
(64) As shown in FIG. **8**, the first lower restriking steels **263** are provided on an upper portion of the second lower die **211**. The first lower restriking steels **263** are connected to the first lower trimming steel **223** and the second lower trimming steel **243**. The first lower restriking steels **263** are configured to support both edge portions of the sunroof hole **71** in the left and right direction thereof. The first lower restriking steels **263** include first undercut forming grooves **267** extending in the downward direction from the upper surface.
(65) As shown in FIG. **9**, the first upper restriking steels **265** are provided on a lower portion of the second upper die **231** in correspondence to the first lower restriking steels **263**. The first upper restriking steels **265** are connected to the first upper trimming steel **225** and the second upper trimming steel **245**. The first upper restriking steels **265** include first undercut forming protrusions **269** extending (e.g., protruding) in an upward direction in correspondence to the first undercut forming grooves **267**.
(66) Referring to FIG. **6**, in various exemplary embodiments of the present disclosure, the first cam forming units **281** are configured to form the hinge mounting portion **79** as shown in FIG. **7** on the rear portion of the first processed plate **30** in a cam forming method. The first cam forming units **281** are provided on the second lower die **211** and the second upper die **231**.
(67) Referring to FIG. **10**, FIG. **11**, and FIG. **12** together with FIG. **6**, the first cam forming units **281** include a swing cam **283**, a driving cylinder **285**, a support block **287**, and a first cam slide **289**.
(68) The swing cam **283** is rotatably provided on a die base **291** mounted on the upper portion of the second lower die **211** in the front and rear direction through a rotation center shaft **293**.
(69) Here, the rotation center shaft **293** is combined to an eccentric point **282** (e.g., a rotation center point) which is biased to the rear side from the center portion of the swing cam **283**. That is, the swing cam **283** may be rotated in the front and rear directions through the rotation center shaft **293** combined to the eccentric point **282**.
(70) The swing cam **283** includes at least one first cam face **284** formed to be inclined from the rear to the front. Furthermore, a first lower cam forming steel **286** is mounted on the swing cam **283**. The first lower cam forming steel **286** is configured to form a lower surface of the hinge mounting portion **79**.
(71) The driving cylinder **285** is provided on the die base **291** to be operable forwards and backwards in correspondence to the swing cam **283**. The driving cylinder **285** may be fixed to the floor of the die base **291** with the swing cam **283** thereon. In one example, the driving cylinder **285** includes an operation rod **288** which is moved forwards and backwards by hydraulic pressure.
(72) The support block **287** is configured to support the lower portion of the front side of the swing cam **283**. The support block **287** is combined to the operation rod **288** of the driving cylinder **285**.
(73) Here, when the operation rod **288** is moved backward by operation of the driving cylinder **285**, the support block **287** is moved backward, and the swing cam **283** may be rotated forward through the rotation center shaft **293**, and when the operation rod **288** moves forward by operation of the driving cylinder **285**, the support block **287** is moved forward, and rotation of the swing cam **283** may be prevented.
(74) The first cam slide **289** is slidably provided on the lower portion of the second upper die **231** in the front and rear directions to be in cam contact with the swing cam **283**. The first cam slide **289** includes at least one second cam face **295** which is in cam contact with the at least one first cam face **284** of the swing cam **283**.
(75) And, an upper cam forming steel **297** corresponding to the first lower cam forming steel **286** of the swing cam **283** is mounted on the first cam slide **289**. The upper cam forming steel **297** is

configured to form the upper surface of the hinge mounting portion **79**.
(76) FIG. **13** is a view exemplarily illustrating a third mold applied to a manufacturing device of a roof panel for a vehicle according to various exemplary embodiments of the present disclosure.
(77) Referring to FIGS. **1** and **13**, in various exemplary embodiments of the present disclosure, the third mold **310** is configured to form the second processed plate **50** (hereinafter see FIG. **7**) processed by the second mold **210** into the predetermined shape.
(78) The third mold **310** flange forms, cam forms, trims, and restriks the second processed plate **50**, and forms the second processed plate **50** into the final roof panel **70** as shown in FIG. **2** and FIG. **3**.
(79) The third mold **310** includes a third lower die **311**, a third upper die **313**, a flange forming unit **321**, a second cam forming unit **331**, a third cam forming unit **341**, a third trimming units **351**, a second restriking unit **361**, and a pair of cam trimming units **371**.
(80) The third lower die **311** is fixed to the floor of a process workshop. The third upper die **313** is provided to be movable in the vertical direction in correspondence to the third lower die **311**. The third upper die **313** may reciprocate in the vertical direction by a hydraulic pressure cylinder known to a person of an ordinary skill in the art.
(81) In various exemplary embodiments of the present disclosure, the flange forming unit **321** is configured to form the down flange **73** as shown in FIG. **2** and FIG. **3** on the edge portion of the sunroof hole **71** (hereinafter see FIG. **7**) of the second processed plate **50** processed by the second mold **210**. The flange forming unit **321** is provided on the third lower die **311** and the third upper die **313**.
(82) The flange forming unit **321** includes a lower flange forming steel **323** and an upper flange forming steel **325** as shown in FIGS. **13** to **15**.
(83) The lower flange forming steel **323** is provided on an upper portion of the third lower die **311**. The third lower flange forming steel **323** is configured to support the edge portion of the sunroof hole **71** of the second processed plate **50**.
(84) Furthermore, the lower flange forming steel **323** includes separating blocks **327** respectively disposed on both sides in the left and right direction thereof. The separating blocks **327** may be provided as portions partially separated from the lower flange forming steel **323**.
(85) The upper flange forming steel **325** is provided on a lower portion of the third upper die **313** in correspondence to the lower flange forming steel **323**. The upper flange forming steel **325** is configured to flange form (e.g., down flange) the edge portion of the sunroof hole **71** placed on the lower flange forming steel **323**.
(86) Here, through holes **329** are formed in both corners of the rear of the upper flange forming steel **325**, and a second undercut forming groove **328** connected to the through holes **329** is formed in each of the separating blocks **327**.
(87) Referring to FIGS. **13** to **15**, in various exemplary embodiments of the present disclosure, the second cam forming unit **331** is configured to form a second undercut face **85** as shown in FIG. **2** and FIG. **3** on at least one corner of the down flange **73** formed by the flange forming unit **321** in a cam forming method. The second cam forming unit **331** is provided on the third lower die **311** and the third upper die **313**.
(88) The second cam forming unit **331** includes a second cam slide **332** and first cam drives **333**.
(89) The second cam slide **332** is slidably provided on an upper portion of the third lower die **311** in a diagonal direction thereof.
(90) A second lower cam forming steel **335** is provided on the second cam slide **332**. The second lower cam forming steel **335** is configured to form the second undercut face **85** on at least one corner of the down flange **73**.
(91) Here, the second lower cam forming steel **335** includes a second undercut forming protrusion **336**. The second undercut forming protrusion **336** may be combined to the second undercut forming groove **328** of the separating block **327** of the lower flange forming steel **323** through the through holes **329** of the upper flange forming steel **325**.

(92) The first cam drives **333** are configured to slide and move the second cam slide **332** in the diagonal direction thereof. The first cam drives **333** are provided on the third lower die **311** and the third upper die **313**.
(93) The first cam drives **333** include a lower drive **333***a* which is slidably provided on the upper portion of the third lower die **311** in the front and rear direction to be in cam contact with the second cam slide **332**. The lower drive **333***a* includes at least one third cam face **334** formed to be inclined from the rear to the front.
(94) Furthermore, the first cam drives **333** include an upper drive **333***b* provided (e.g., fixed) on the lower portion of the third upper die **311** to be in cam contact with the lower drive **333***a*. The upper drive **333***b* includes at least one fourth cam face **337** which is in cam contact with the at least one third cam surface **334** of the lower drive **333***a*.
(95) FIG. **16** and FIG. **17** are views exemplarily illustrating a third cam forming unit of a third mold applied to a manufacturing device of a roof panel for a vehicle according to various exemplary embodiments of the present disclosure.
(96) Referring to FIGS. **16** and **17**, in various exemplary embodiments of the present disclosure, the third cam forming unit **341** is configured to form the up flange **77** as shown in FIG. **2** and FIG. **3** in a cam forming method on the at least one molding mounting portion **75** (hereinafter see FIG. **7**) formed on the second processed plate **50** (hereinafter see FIG. **7**). The third cam forming unit **341** is provided on the third lower die **311** and the third upper die **313**.
(97) The third cam forming unit **341** includes third cam slides **342**, a plurality of seesaw cams **343**, a second cam drive **344**, and a plurality of third cam drives **345**.
(98) The third cam slides **342** are slidably provided on an upper portion of the third lower die **311** in the left and right direction thereof. The third cam slides **342** are disposed on both left and right sides of the third lower die **311**, and are combined to the respective separating blocks **327** of the lower flange forming steel **323**.
(99) The plurality of seesaw cams **343** are fixed to both sides of the third lower die **311** in the left and right direction, and are mounted on the third cam slide **342**. Each of the seesaw cams **343** includes a cam block **347** which is rotated in a seesaw type in the left and right direction in the mounting block **346**.
(100) Here, a third lower cam forming steel **348** is provided on the cam block **347**. The third lower cam forming steel **348** is configured to form the up flange **77**.
(101) The second cam drive **344** is configured to slide and move the third cam slide **342**. The second cam drive **344** is provided on a lower portion of the third upper die **313** to be in cam contact with the third cam slide **342**.
(102) And, the plurality of third cam drives **345** are configured to rotate the respective cam blocks **347** of the plurality of seesaw cams **343**. The plurality of third cam drives **345** are provided in a lower portion of the third upper die **313** to be in cam contact with the respective cam blocks **347**.
(103) FIG. **18**, FIG. **19** and FIG. **20** are views exemplarily illustrating a third forming unit, a second restriking unit, and a cam trimming unit of a third mold applied to a manufacturing device of a roof panel for a vehicle according to various exemplary embodiments of the present disclosure.
(104) Referring to FIGS. **18** to **20**, in various exemplary embodiments of the present disclosure, the third mold **310** further includes an upper pad **315** elastically supported on a lower portion of the third upper die **313**.
(105) The upper pad **315** is elastically supported on the lower portion of the third upper die **313** through a plurality of gas springs **317** and is provided to be movable in the vertical direction thereof.
(106) In various exemplary embodiments of the present disclosure, the third trimming unit **351** is configured to trim the rear end portion (e.g., the rear edge portion) of the second processed plate **50** (hereinafter see FIG. **7**). The third trimming unit **351** is provided on the third lower die **311** and the third upper die **313**.

(107) The third trimming unit **351** includes a third lower trimming steel **353** and a third upper trimming steel **355**.
(108) The third lower trimming steel **353** is provided on an upper portion of the third lower die **311**. The third lower trimming steel **353** is configured to support the edge portion of the rear portion of the second processed plate **50**.
(109) The third upper trimming steel **355** is provided on a lower portion of the upper pad **315** in correspondence to the third lower trimming steel **353**. The third upper trimming steel **355** is configured to cut the rear end portion of the second processed plate **50** placed on the third lower trimming steel **353**.
(110) In various exemplary embodiments of the present disclosure, the second restriking unit **361** is configured to form the water receiving portion **81** connected to the hinge mounting portion **79** as shown in FIG. **2** and FIG. **3** in the rear portion of the second processed plate **50**. The second restriking unit **361** is provided on the third lower die **311** and the third upper die **313**.
(111) Here, after trimming the rear end portion of the second processed plate **50** through the third trimming unit **351**, the second restriking unit **361** may form the water receiving portion **81**.
(112) The second restriking unit **361** includes a second lower restriking steel **363** and a second upper restriking steel **365**.
(113) The second lower restriking steel **363** is provided on the third lower die **311**. The second lower restriking steel **363** is connected to the third lower trimming steel **353** of the third trimming unit **351**. The second lower restriking steel **363** is configured to support the rear portion of the second processed plate **50**.
(114) The second upper restriking steel **365** is connected to the third upper trimming steel **355** of the third trimming unit **351**. The second upper restriking steel **365** is provided on a lower portion of the upper pad **315** with a height difference from the third upper trimming steel **355** in correspondence to the second lower restriking steel **363**. Here, the second upper restriking steel **365** may be disposed at a higher position than that of the third upper trimming steel **355**.
(115) In various exemplary embodiments of the present disclosure, the pair of cam trimming units **371** are configured to respectively trim both sides of the rear end portion of the second processed plate **50** in a cam trimming method. The pair of cam trimming units **371** are provided on the third lower die **311** and the third upper die **313**.
(116) Here, the pair of cam trimming units **371** may respectively trim both sides of the rear end portion of the second processed plate **50** after forming the water receiving portion **81** through the second restriking unit **361**.
(117) Each of the pair of cam trimming units **371** includes a fourth cam slide **373** and a fourth cam drive **375**
(118) The fourth cam slide **373** is slidably provided on a lower portion of the third upper die **313** in the front and rear direction thereof. The fourth cam slide **373** includes at least one fifth cam face **377** formed to be inclined from the front to the rear.
(119) Furthermore, a cam trimming steel **379** is provided on the fourth cam slide **373**. The cam trimming steel **379** is configured to cut both sides of the rear end portion of the second processed plate **50**.
(120) The fourth cam drive **375** is configured to slide and move the fourth cam slide **373**. The fourth cam drive **375** is provided on an upper portion of the third lower die **311**.
(121) The fourth cam drive **375** includes at least one sixth cam face **381** which is in cam contact with the at least one fifth cam face **377** of the fourth cam slide **373**.
(122) Hereinafter, the operation of the manufacturing device **100** of the roof panel for the vehicle according to various exemplary embodiments of the present disclosure configured as described above will be described in detail with reference to FIGS. **1** to **20**.
(123) First, in a state in which the first upper die **117** of the first mold **110** is moved in the upward direction, the blank **10** is loaded onto the blank holder **113** and the lower drawing steel **115** of the

first lower die **111**.

(124) In the present state, when the first upper die **117** moves in the downward direction, the blank holder **113** supports the edge portion of the blank **10** and moves in the downward direction by the first upper die **117**. Accordingly, the upper drawing steel **119** of the first upper die **117** presses the blank **10**.

(125) Accordingly, the first mold **110** draws the blank **10** into a predetermined shape by the blank holder **113**, the lower drawing steel **115**, and the upper drawing steel **119**.

(126) Next, in a state in which the second upper die **231** of the second mold **210** is moved in the upward direction, the first processed plate **30** processed by the first mold **110** is loaded onto the second lower die **211**.

(127) In the present state, when the second upper die **231** moves in the downward direction, the first lower trimming steel **223** and the first upper trimming steel **225** of the first trimming unit **221** entirely trim the edge portion of the front portion of the first processed plate **30** and both sides thereof in the left and right direction thereof.

(128) Here, the first lower trimming steel **223** and the first upper trimming steel **225** trim both edge portions of the first processed plate **30** in the left and right direction, and form the at least one molding mounting portion **75** on each of both edge portions.

(129) In the present process, the second lower trimming steel **243** and the second upper trimming steel **245** of the second trimming unit **241** trim the sunroof hole **71** in the front portion of the first processed plate **30**.

(130) In this regard, the second upper trimming steel **245** may trim the sunroof hole **71** while first pressing the first processed plate **30** through the pressing protrusion **247**. Accordingly, the second trimming unit **241** may minimize the occurrence of a burr formed on the edge portion of the sunroof hole **71** by the aerial trimming of the second upper trimming steel **245**.

(131) Simultaneously, the first lower restriking steels **263** and the first upper restriking steels **265** of the first restriking unit **261** respectively form the first undercut faces **83** on both edge portions of the sunroof hole **71** in the left and right direction thereof.

(132) Here, the first undercut faces **83** may be formed while the first undercut forming protrusions **269** of the first upper restriking steels **265** are combined to the first undercut forming grooves **267** of the first lower restriking steels **263**.

(133) In the present process, the swing cam **283** and the first cam slide **289** of the first cam forming units **281** form the hinge mounting portion **79** on the rear portion of the first processed plate **30** in a cam forming method.

(134) In the above process, in a state in which the operation rod **288** is moved forward by operation of the driving cylinder **285**, the support block **287** supports the lower portion of the front side of the swing cam **283**. Accordingly, the swing cam **283** is maintained in a horizontal state by the support block **287**.

(135) Accordingly, the first cam slide **289** in cam contact with the swing cam **283** slides and moves, and the first lower cam forming steel **286** of the swing cam **283** and the upper cam forming steel **297** of the first cam slide **289** may form the hinge mounting portion **79**.

(136) Here, as the first cam slide **289** in cam contact with the swing cam **283** forms the hinge mounting portion **79** from the rear side as described above, the second mold **210** may smoothly exhaust a scrap cut by the first trimming unit **221**.

(137) Furthermore, after completing a trimming process, a restriking process, and a cam forming process of the first processed plate **30** as described above, when the second upper die **231** moves in the upward direction, the operation rod **288** moves backward by operation of the driving cylinder **285**, and accordingly, the support block **287** moves backward thereof.

(138) Accordingly, the swing cam **283** rotates forward with respect to an eccentric point **282** via the rotation center shaft **293**. Accordingly, the swing cam **283** may smoothly eject the second processed plate **50** on which the processing of the first processed plate **30** is completed while avoiding the

interference of the hinge mounting portion **79**.

(139) Next, in a state in which the third upper die **313** of the third mold **310** is moved in the upward direction, the second processed plate **50** processed by the second mold **210** is loaded onto the third lower die **311**.

(140) In the present state, when the third upper die **313** moves in the downward direction, the lower flange forming steel **323** and the upper flange forming steel **325** of the flange forming unit **321** form the down flange **73** on the edge portion of the sunroof hole **71** of the second processed plate **50**.

(141) Simultaneously, the second cam slide **332** and the first cam drives **333** of the second cam forming unit **331** form the second undercut faces **85** on at least one corner of the down flange **73** in a cam forming method.

(142) In the above process, while the upper drive **333***b* of the first cam drives **333** is in cam contact with the lower drive **333***a*, the lower drive **333***a* slides backward thereof. Accordingly, the second cam slide **332** in cam contact with the lower drive **333***a* slides and moves (e.g., moves forward) in a diagonal direction thereof.

(143) Accordingly, the second lower cam forming steel **335** of the second cam slide **332** is combined to the second undercut forming groove **328** of the separating block **327** of the lower flange forming steel **323** through the through hole **329** of the upper flange forming steel **325**.

(144) Accordingly, the second lower cam forming steel **335** may form the second undercut faces **85** at the at least one corner of the down flange **73** through the second undercut forming protrusion **336**.

(145) In the present process, the third cam slide **342**, the plurality of seesaw cams **343**, the second cam drive **344**, and the plurality of third cam drives **345** of the third cam forming unit **341** form the up flange **77** on the at least one molding mounting portion **75** in a cam forming method.

(146) In the above process, the second cam drive **344** is in cam contact with the third cam slide **342**, the third cam slide **342** slides and moves, and each of the separating blocks **327** of the lower flange forming steel **323** is in a forward moved state.

(147) In the present state, each of the third cam drives **345** in cam contact with the cam block **347** of each of the seesaw cams **343** rotates the cam block **34**, and may form the up flange **77** on the at least one molding mounting portion **75** through the third lower cam forming steel **348** of the cam block **347**.

(148) Simultaneously, the third lower trimming steel **353** and the third upper trimming steel **355** of the third trimming unit **351** preferentially trim the rear end portion of the second processed plate **50**.

(149) Accordingly, the second lower restriking steel **363** and the second upper restriking steel **365** of the second restriking unit **361** form the water receiving portion **81** connected to the hinge mounting portion **79** on the rear portion of the second processed plate **50**.

(150) Next, after completing the trimming process and the restriking process as described above, the third upper die **313** continuously descends while pressing the upper pad **315** through the plurality of gas springs **317**.

(151) At the present time, the fourth cam slide **373** and the fourth cam drive **375** of the pair of cam trimming units **371** trim both rear end portions of the second processed plate in a cam trimming method.

(152) Here, the fourth cam drive **375** is in cam contact with the fourth cam slide **373**, the fourth cam slide **373** slides and moves, and the cam trimming steel **379** of the fourth cam slide **373** may trim each of both sides of the rear end portion of the second processed plate **50**.

(153) Furthermore, after completing the series of processes as described above, when the third upper die **313** moves in the upward direction, the second cam slide **332** of the second cam forming unit **331** returns to its original position thereof. That is, the second lower cam forming steel **335** of the second cam slide **332** returns to its original position (e.g., moves backward) through the

through hole **329** of the upper flange forming steel **325**.
(154) Accordingly, the third cam slide **342** of the third cam forming unit **341** returns (e.g., moves backward) to its original position together with each of the separating blocks **327** of the lower flange forming steel **323**.
(155) Therefore, the lower flange forming steel **323** of the flange forming unit **321** may smoothly eject the roof panel **70** of which processing has been completed while avoiding the interference of the second undercut face **85** formed on the at least one corner of the down flange **73**.
(156) Unlike the related art, the manufacturing device **100** of the roof panel for the vehicle according to various exemplary embodiments of the present disclosure as described so far may manufacture the roof panel **70** of the final finished product while performing the series of processes as described above by use of the first mold **110**, the second mold **210**, and the third mold **310**.
(157) Therefore, the manufacturing device **100** of the roof panel for the vehicle according to various exemplary embodiments of the present disclosure may reduce the number of molds and the number of processes required for manufacturing the roof panel **70**. Accordingly, the mold investment cost, the mold material cost, the processing cost, and the cost required for production management may be reduced, and the actual cost competitiveness may be improved.
(158) Although exemplary embodiments of the present disclosure have been described above, the present disclosure is not limited thereto, and it is obvious that various modifications may be made within the scope of the claims, the detailed description of the invention, and the accompanying drawings and that also fall within the scope of the present disclosure.
(159) For convenience in explanation and accurate definition in the appended claims, the terms "upper", "lower", "inner", "outer", "up", "down", "upwards", "downwards", "front", "rear", "back", "inside", "outside", "inwardly", "outwardly", "interior", "exterior", "internal", "external", "forwards", and "backwards" are used to describe features of the exemplary embodiments with reference to the positions of such features as displayed in the figures. It will be further understood that the term "connect" or its derivatives refer both to direct and indirect connection.
(160) The term "and/or" may include a combination of a plurality of related listed items or any of a plurality of related listed items. For example, "A and/or B" includes all three cases such as "A", "B", and "A and B".
(161) The foregoing descriptions of specific exemplary embodiments of the present disclosure have been presented for purposes of illustration and description. They are not intended to be exhaustive or to limit the present disclosure to the precise forms disclosed, and obviously many modifications and variations are possible in light of the above teachings. The exemplary embodiments were chosen and described in order to explain certain principles of the invention and their practical application, to enable others skilled in the art to make and utilize various exemplary embodiments of the present disclosure, as well as various alternatives and modifications thereof. It is intended that the scope of the present disclosure be defined by the Claims appended hereto and their equivalents.

Claims

1. A manufacturing apparatus of a roof panel for a vehicle, the manufacturing apparatus comprising: a first mold in which a blank holder and a lower drawing steel are provided in a first lower die, and an upper drawing steel corresponding to the lower drawing steel is provided in a first upper die, to draw a blank in a predetermined shape; a second mold including first trimming units provided in a second lower die and a second upper die to trim an edge portion of a first processed plate processed by the first mold, second trimming units provided on the second lower die and the second upper die to trim a sunroof hole in a front portion of the first processed plate, first restriking units provided on the second lower die and the second upper die to form first undercut faces on first and second edge portions of the sunroof hole in a left and right direction, and first cam forming

units provided on the second lower die and the second upper die to form a hinge mounting portion on a rear portion of the first processed plate; and a third mold including flange forming units provided in a third lower die and a third upper die to form a down flange on an edge portion of the sunroof hole of a second processed plate processed by the second mold, and second restriking units provided in the third lower die and the third upper die to form a water receiving portion connected to the hinge mounting portion on a rear portion of the second processed plate.

2. The manufacturing apparatus of claim 1, wherein the first trimming units form at least one molding mounting portion on each of first and second edge portions of the first processed plate in the left and right direction.

3. The manufacturing apparatus of claim 1, wherein the third mold further includes: third trimming units provided on the third lower die and the third upper die to trim a rear end portion of the second processed plate; and a pair of cam trimming units provided on the third lower die and the third upper die to trim first and second sides of the rear end portion of the second processed plate, respectively.

4. The manufacturing apparatus of claim 3, wherein the third mold trims the rear end portion of the second processed plate through the third trimming units before forming the water receiving portion through the second restriking units, and wherein the third mold trims first and second sides of the rear end portion of the second processed plate, respectively, through the pair of cam trimming units after forming the water receiving portion through the second restriking units.

5. The manufacturing apparatus of claim 3, wherein the third mold further includes second cam forming units provided on the third lower die and the third upper die to form a second undercut face on at least one corner of the down flange.

6. The manufacturing apparatus of claim 5, wherein the first trimming unit forms at least one molding mounting portion on each of first and second edge portions of the first processed plate in the left and right direction, and wherein the third mold further includes third cam forming units provided on the third lower die and the third upper die, to form an up flange on the at least one molding mounting portion formed on the second processed plate.

7. The manufacturing apparatus of claim 6, wherein the flange forming units include: a lower flange forming steel including separating blocks respectively disposed on first and second sides thereof in the left and right directions, and provided on the third lower die; and an upper flange forming steel provided in the third upper die in correspondence to the lower flange forming steel, and including through holes formed in first and second corners of the rear thereof.

8. The manufacturing apparatus of claim 7, wherein each of the separating blocks includes a second undercut forming groove connected to each of the through holes.

9. The manufacturing apparatus of claim 8, wherein the second cam forming units include: a second cam slide including a second lower cam forming steel including second undercut forming protrusions combined to the second undercut forming grooves through the through holes, the second cam slide being slidably provided on the third lower die; and first cam drives provided on the third lower die and the third upper die to slide and move the second cam slide.

10. The manufacturing apparatus of claim 7, wherein the third cam forming units include: a third cam slide combined to each of the separating blocks and slidably provided on the third lower die; a plurality of seesaw cams fixed to first and second sides of the third lower die in the left and right direction, mounted on the third cam slide, and each including a third lower cam forming steel provided thereon; a second cam drive provided on the third upper die to be in cam contact with the third cam slide; and a plurality of third cam drives provided on the third upper die to be in cam contact with each of the seesaw cams.

11. The manufacturing apparatus of claim 3, wherein the third mold further includes an upper pad movably provided on the third upper die in a vertical direction through a plurality of gas springs.

12. The manufacturing apparatus of claim 11, wherein the third trimming unit includes: a third lower trimming steel provided on the third lower die; and a third upper trimming steel provided on

the upper pad in correspondence to the third lower trimming steel.

13. The manufacturing apparatus of claim 12, wherein the second restriking units include: a second lower restriking steel connected to the third lower trimming steel; and a second upper restriking steel provided on the upper pad with a height difference from the third upper trimming steel.

14. The manufacturing apparatus of claim 11, wherein each of the pair of cam trimming units includes: a fourth cam slide slidably provided on the third upper die and including a cam trimming steel provided thereon; and a fourth cam drive provided on the third lower die to be in cam contact with the fourth cam slide.

15. The manufacturing apparatus of claim 1, wherein the first trimming units include: a first lower trimming steel provided on the second lower die; and a first upper trimming steel provided on the second upper die in correspondence to the first lower trimming steel.

16. The manufacturing apparatus of claim 15, wherein the second trimming units include: a second lower trimming steel connected to an internal edge portion of the first lower trimming steel; and a second upper trimming steel connected to an internal edge portion of the first upper trimming steel in correspondence to the second lower trimming steel, and including a pressing protrusion extending in a downward direction to support the first processed plate.

17. The manufacturing apparatus of claim 16, wherein the first restriking units include: first lower restriking steels connected to the first lower trimming steel and the second lower trimming steel and including first undercut forming grooves formed therein; and first upper restriking steels connected to the first upper trimming steel and the second upper trimming steel and including first undercut forming protrusions corresponding to the first undercut forming grooves formed thereon.

18. The manufacturing apparatus of claim 1, wherein the first cam forming units include: a swing cam rotatably provided on a die base mounted on the second lower die through a rotation center shaft in a front and rear direction and including a first lower cam forming steel mounted thereon; and a first cam slide slidably provided on the second upper die to be in cam contact with the swing cam, and including an upper cam forming steel corresponding to the first lower cam forming steel mounted thereon.

19. The manufacturing apparatus of claim 18, wherein the rotation center shaft is combined to an eccentric point biased to a rear side from a center portion of the swing cam.

20. The manufacturing apparatus of claim 19, wherein the first cam forming units further include: a driving cylinder provided on the die base to be operable forwards and backwards in the front and rear direction thereof; and a support block combined to an operation rod of the driving cylinder to support a front side of the swing cam.

21. A manufacturing apparatus of a roof panel for a vehicle, the manufacturing apparatus including: a first mold in which a blank holder and a lower drawing steel are provided in a first lower die, and an upper drawing steel corresponding to the lower drawing steel is provided in a first upper die, to draw a blank in a predetermined shape; a second mold including first trimming units provided in a second lower die and a second upper die to trim an edge portion of a first processed plate processed by the first mold, second trimming units provided on the second lower die and the second upper die to trim a sunroof hole in a front portion of the first processed plate, and first restriking units provided on the second lower die and the second upper die to form first undercut faces on first and second edge portions of the sunroof hole in a left and right direction thereof; and a third mold including flange forming units provided in a third lower die and a third upper die to form a down flange on an edge portion of the sunroof hole of a second processed plate processed by the second mold.

22. A manufacturing apparatus of a roof panel for a vehicle, the manufacturing apparatus including: a first mold in which a blank holder and a lower drawing steel are provided in a first lower die, and an upper drawing steel corresponding to the lower drawing steel is provided in a first upper die, to draw a blank in a predetermined shape; a second mold including first trimming units provided in a second lower die and a second upper die to trim an edge portion of a first processed plate processed

by the first mold, and first cam forming units provided on the second lower die and the second upper die to form a hinge mounting portion on a rear portion of the first processed plate; and a third mold including second restriking units provided in the third lower die and the third upper die to form a water receiving portion connected to the hinge mounting portion on a rear portion of a second processed plate processed by the second mold.